

## ISTARVSHAN, Tajikistan

WMO: 387130

Lat: 39.900N

Lon: 68.983E

Elev: 1005

StdP: 89.82

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.5      | -9.2 | -16.9                         | 1.0 | -10.8 | -14.7 | 1.2 | -7.4 | 4.5                   | 3.1  | 4.3 | 2.2  | 1.7                   | 140  | 0.422 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.0                   | 34.3            | 18.0 | 33.0 | 17.2 | 31.6 | 16.9 | 19.3                | 32.0 | 18.5 | 30.6 | 17.8 | 29.3 | 2.2                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 14.7                            | 11.8 | 23.0 | 13.9 | 11.2 | 23.0 | 13.1 | 10.6 | 22.9 | 59.2          | 32.3 | 56.3 | 30.6 | 54.1 | 29.5 | 21.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 4.8           | 4.1  | 3.4 | -13.8                      | 36.6 | 3.3                | 1.7 | -16.2                                              | 37.8 | -18.2      | 38.8 | -20.1      | 39.7 | -22.5      | 40.9 |     |     |
| (5)               |      |     | -14.4                      | 20.5 | 3.3                | 0.9 | -16.8                                              | 21.2 | -18.7      | 21.7 | -20.6      | 22.2 | -22.9      | 22.9 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.2   | 0.0   | 1.0   | 6.3   | 11.7  | 17.5  | 22.8  | 25.0  | 23.2  | 18.6  | 11.7  | 6.1   | 1.9   |  |  |
|                                                                        | DBStd       | 9.69   | 4.06  | 5.17  | 4.85  | 4.33  | 3.70  | 2.90  | 2.58  | 2.81  | 3.63  | 4.85  | 5.02  | 4.81  |  |  |
|                                                                        | HDD10.0     | 1162   | 309   | 255   | 134   | 31    | 4     | 0     | 0     | 0     | 2     | 39    | 134   | 254   |  |  |
|                                                                        | HDD18.3     | 2818   | 568   | 486   | 373   | 202   | 58    | 3     | 0     | 1     | 39    | 209   | 368   | 511   |  |  |
|                                                                        | CDD10.0     | 1967   | 0     | 2     | 19    | 81    | 236   | 384   | 466   | 410   | 259   | 92    | 16    | 2     |  |  |
|                                                                        | CDD18.3     | 582    | 0     | 0     | 0     | 2     | 32    | 137   | 207   | 153   | 46    | 4     | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 6373   | 0     | 0     | 1     | 29    | 313   | 1439  | 2406  | 1611  | 522   | 48    | 4     | 0     |  |  |
|                                                                        | CDH26.7     | 2574   | 0     | 0     | 0     | 3     | 65    | 563   | 1122  | 666   | 148   | 7     | 1     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.8    | 1.7   | 1.8   | 2.0   | 2.0   | 2.0   | 2.0   | 2.0   | 1.9   | 1.8   | 1.7   | 1.6   | 1.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 519    | 54    | 71    | 84    | 80    | 60    | 25    | 14    | 7     | 7     | 23    | 42    | 52    |  |  |
|                                                                        | PrecMax     | 736    | 97    | 121   | 135   | 187   | 120   | 54    | 37    | 21    | 19    | 50    | 112   | 100   |  |  |
|                                                                        | PrecMin     | 326    | 22    | 23    | 24    | 26    | 11    | 9     | 5     | 3     | 1     | 4     | 12    | 7     |  |  |
|                                                                        | PrecStd     | 101    | 21    | 23    | 27    | 31    | 29    | 11    | 7     | 4     | 4     | 11    | 25    | 25    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 12.5  | 16.9  | 21.2  | 26.9  | 30.6  | 34.6  | 36.7  | 34.8  | 31.7  | 27.7  | 21.7  | 16.0  |  |  |
|                                                                        |             | MCWB   | 5.6   | 8.6   | 11.5  | 15.7  | 16.6  | 17.4  | 19.1  | 18.4  | 15.4  | 14.8  | 10.7  | 7.2   |  |  |
|                                                                        | 2%          | DB     | 9.8   | 12.8  | 18.5  | 23.6  | 28.5  | 32.9  | 35.2  | 33.5  | 30.1  | 24.3  | 18.2  | 13.1  |  |  |
|                                                                        |             | MCWB   | 3.9   | 6.4   | 10.5  | 13.6  | 16.3  | 16.7  | 18.3  | 17.8  | 15.3  | 12.8  | 9.0   | 6.5   |  |  |
|                                                                        | 5%          | DB     | 7.7   | 9.9   | 16.3  | 21.2  | 26.7  | 31.6  | 33.6  | 32.0  | 28.5  | 22.0  | 15.7  | 10.7  |  |  |
|                                                                        |             | MCWB   | 2.9   | 4.9   | 9.9   | 12.8  | 15.7  | 16.7  | 17.8  | 16.6  | 15.2  | 11.8  | 8.1   | 5.1   |  |  |
|                                                                        | 10%         | DB     | 5.6   | 8.0   | 13.7  | 19.0  | 24.9  | 30.1  | 32.1  | 30.5  | 26.6  | 19.9  | 13.5  | 8.4   |  |  |
|                                                                        |             | MCWB   | 1.7   | 3.8   | 8.1   | 12.2  | 15.3  | 16.3  | 17.5  | 16.4  | 14.5  | 10.8  | 7.3   | 3.9   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.0   | 8.9   | 12.6  | 16.0  | 18.1  | 19.2  | 20.4  | 20.0  | 16.8  | 14.9  | 12.0  | 8.4   |  |  |
|                                                                        |             | MCDB   | 11.6  | 16.0  | 19.1  | 23.6  | 27.1  | 31.1  | 34.2  | 32.7  | 28.6  | 26.9  | 18.9  | 14.3  |  |  |
|                                                                        | 2%          | WB     | 4.5   | 7.1   | 11.2  | 14.7  | 17.2  | 18.3  | 19.6  | 18.7  | 15.8  | 13.3  | 10.1  | 6.8   |  |  |
|                                                                        |             | MCDB   | 8.9   | 11.6  | 17.4  | 22.0  | 25.6  | 29.6  | 32.9  | 31.5  | 27.6  | 22.9  | 15.8  | 12.0  |  |  |
|                                                                        | 5%          | WB     | 3.2   | 5.5   | 10.0  | 13.5  | 16.4  | 17.7  | 18.9  | 17.8  | 15.3  | 12.3  | 9.0   | 5.5   |  |  |
|                                                                        |             | MCDB   | 7.1   | 9.4   | 15.6  | 19.8  | 24.4  | 28.6  | 31.6  | 30.1  | 27.3  | 21.0  | 14.7  | 9.5   |  |  |
|                                                                        | 10%         | WB     | 2.1   | 4.1   | 8.7   | 12.6  | 15.7  | 17.1  | 18.1  | 16.9  | 14.8  | 11.1  | 7.8   | 4.4   |  |  |
|                                                                        |             | MCDB   | 5.0   | 7.5   | 13.2  | 18.0  | 23.4  | 27.5  | 30.3  | 28.5  | 26.4  | 18.9  | 12.9  | 8.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.5   | 8.2   | 8.5   | 9.3   | 10.4  | 11.7  | 12.0  | 12.4  | 12.5  | 11.3  | 9.4   | 8.4   |  |  |
|                                                                        |             | MCDBR  | 10.6  | 10.5  | 11.8  | 12.4  | 13.0  | 13.5  | 13.7  | 13.6  | 14.8  | 13.8  | 12.8  | 11.0  |  |  |
|                                                                        | 5% WB       | MCWBR  | 7.2   | 7.2   | 7.0   | 6.8   | 6.3   | 5.5   | 5.8   | 5.6   | 6.7   | 7.1   | 6.8   | 6.7   |  |  |
|                                                                        |             | MCWBR  | 9.5   | 9.8   | 10.6  | 11.0  | 11.5  | 12.2  | 12.7  | 13.1  | 14.0  | 12.9  | 11.2  | 9.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.341 | 0.388 | 0.454 | 0.474 | 0.474 | 0.474 | 0.464 | 0.437 | 0.418 | 0.446 | 0.404 | 0.343 |  |  |
|                                                                        | taud        |        | 2.177 | 2.053 | 1.929 | 1.929 | 1.983 | 1.988 | 2.015 | 2.080 | 2.074 | 1.939 | 2.016 | 2.154 |  |  |
|                                                                        | Ebn at Noon |        | 796   | 804   | 788   | 803   | 812   | 812   | 815   | 828   | 815   | 725   | 701   | 755   |  |  |
|                                                                        | Edh at Noon |        | 104   | 136   | 172   | 184   | 178   | 177   | 171   | 156   | 147   | 151   | 122   | 99    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.00  | 2.76  | 3.81  | 5.02  | 6.26  | 7.32  | 7.43  | 6.82  | 5.61  | 3.70  | 2.23  | 1.74  |  |  |
|                                                                        | RadStd      |        | 0.20  | 0.36  | 0.39  | 0.59  | 0.44  | 0.32  | 0.31  | 0.34  | 0.21  | 0.37  | 0.31  | 0.23  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (12 neighbors) | N/A   | N/A     | N/A    | +0.52   | N/A   | N/A   | N/A         | N/A     | +93     | +67     |

Nomenclature: See separate page